

Atomic, Molecular and Optical (AMO) Physics

Controlling, tracking and visualizing matter through the help of radiation

Fred. Ongonwou, Department of Engineering Science, Morishita Laboratory, The University of Electro-Communications

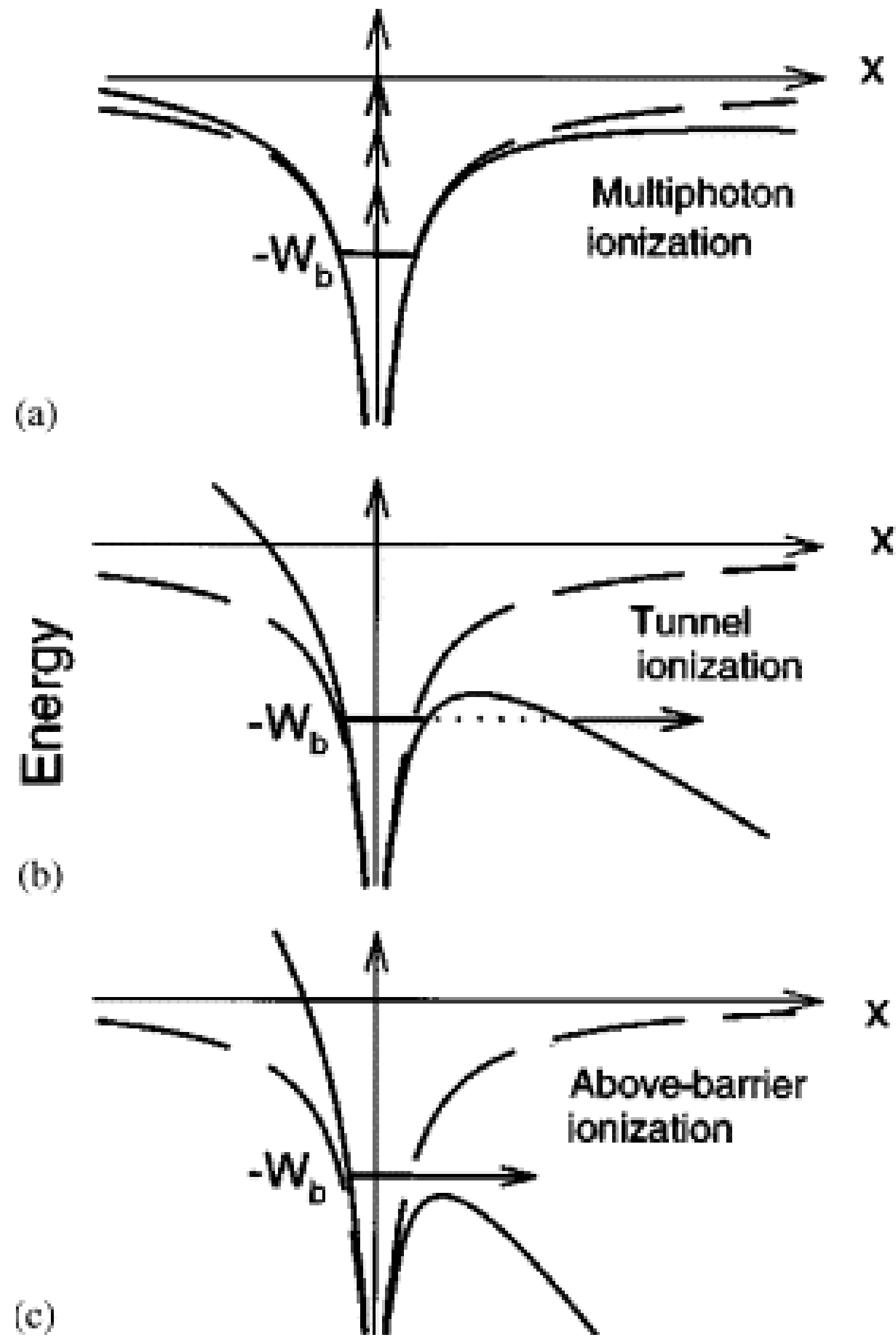
Atomic, Molecular and Optical (AMO) physics explores how atoms and molecules interact with light. Using powerful lasers, we can control electrons, generate new type of light, and even take pictures of matter in motion – at timescales a billion times faster than a camera flash. These techniques let us probe the structure and dynamics of molecules, understand fundamental quantum processes, and build technologies like quantum sensors, ultrafast microscopes and new light sources. AMO physics is the science behind some of the most precise measurements and fastest tools ever made.

Classical physics (High school)

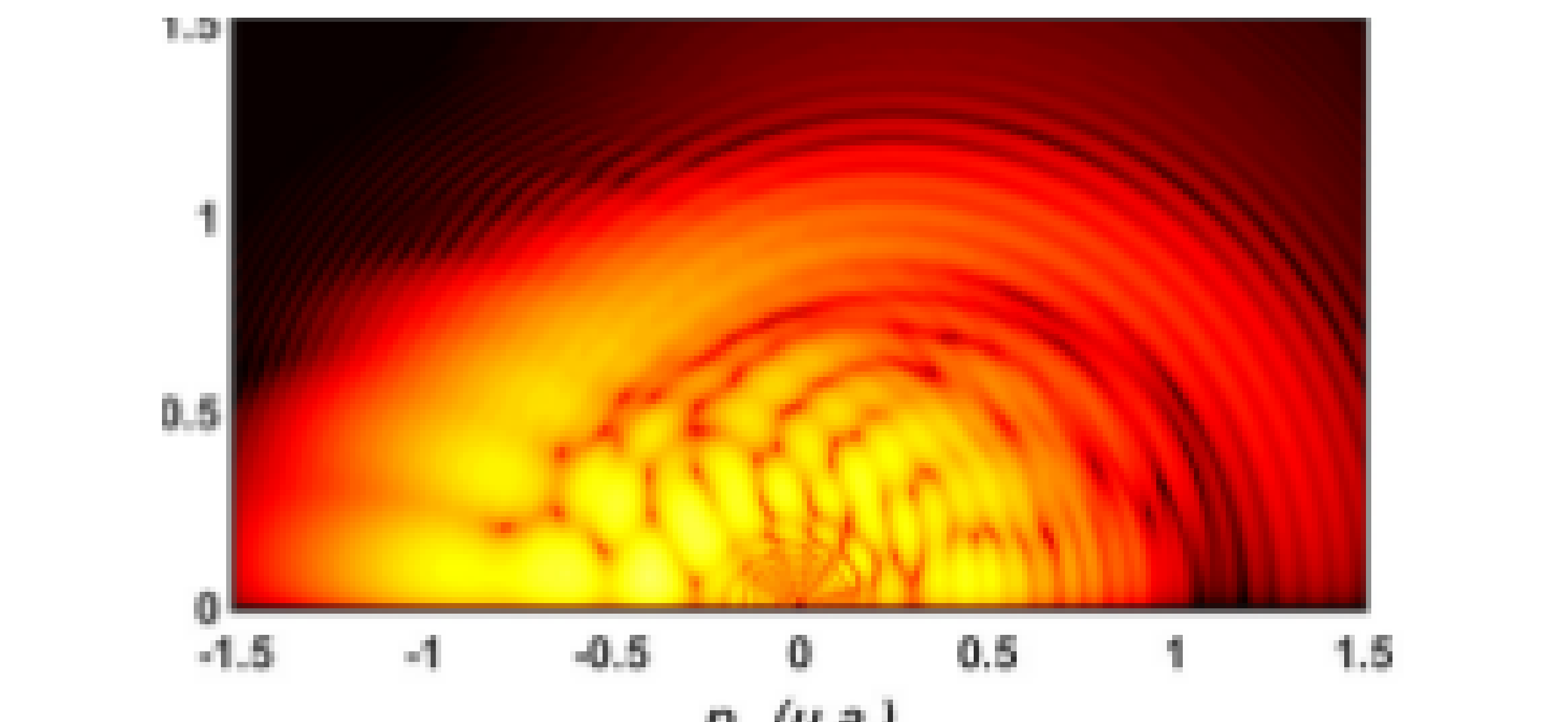
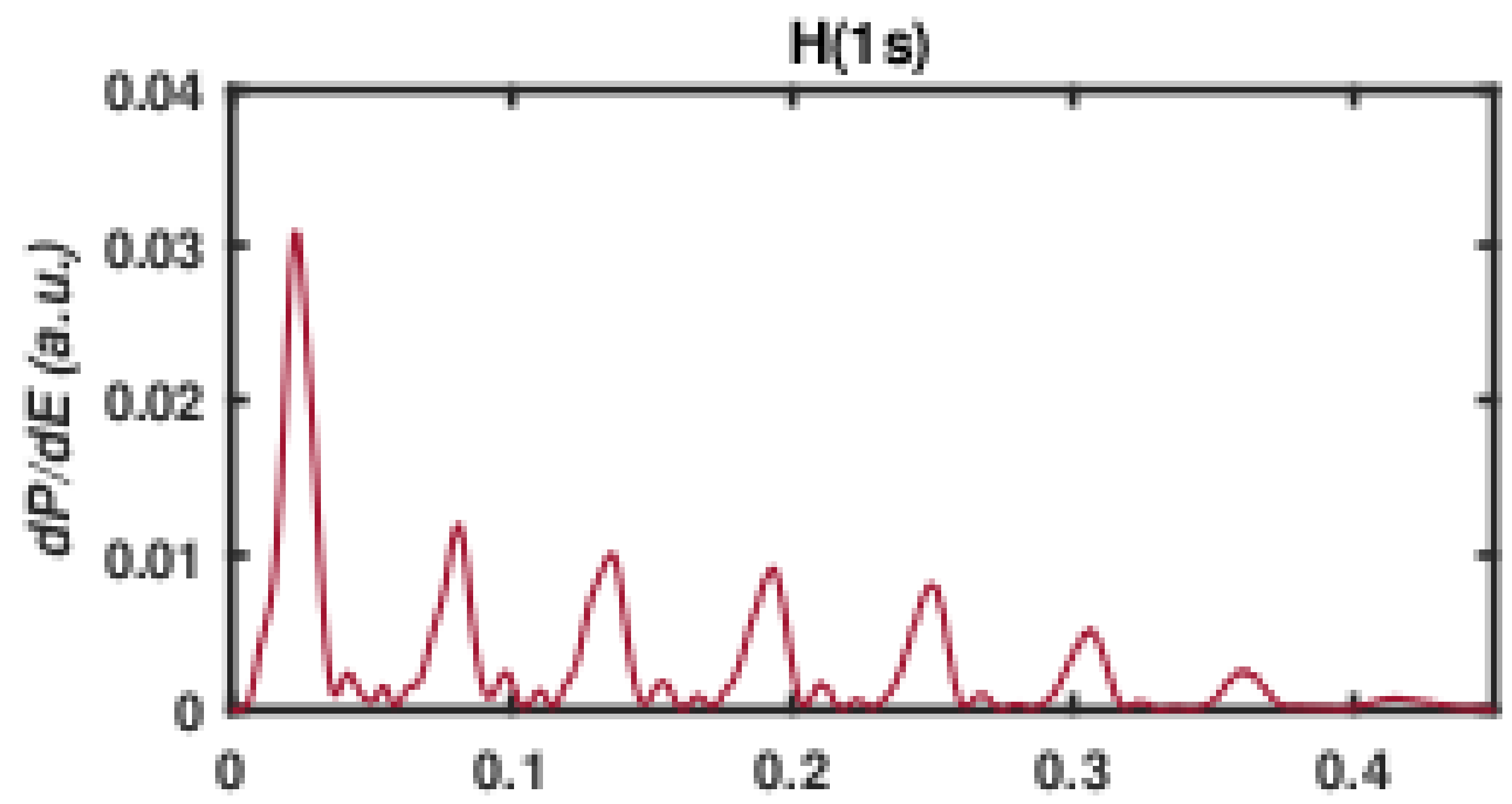
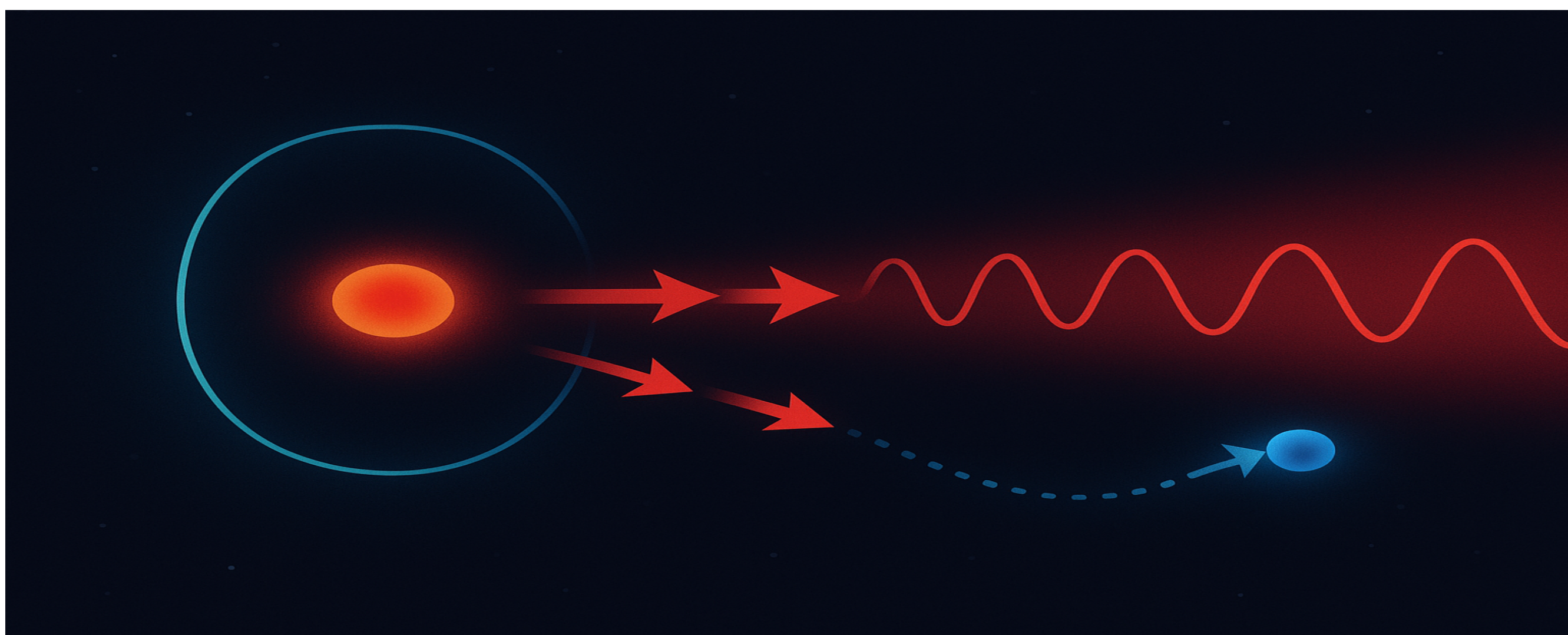
$$E = \frac{1}{2}mv^2 - \frac{e^2}{4\pi\epsilon_0 x} - e\mathbf{E} \cdot \mathbf{r} \quad \longrightarrow \quad i\frac{\partial}{\partial t}\psi(\mathbf{r}, t) = \left(-i\hbar\frac{\partial^2}{\partial r^2} - \frac{e^2}{4\pi\epsilon_0 x} - e\mathbf{E} \cdot \mathbf{r} \right)\Psi(x, t)$$

Quantum physics (University)

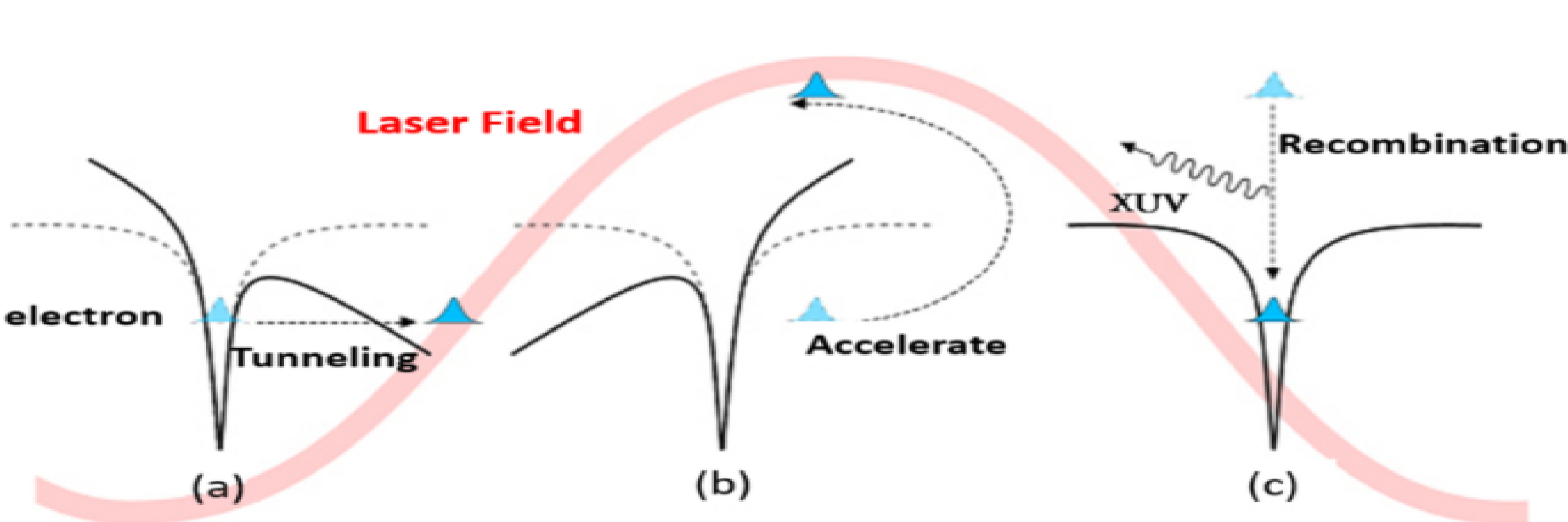
1. Ionizations



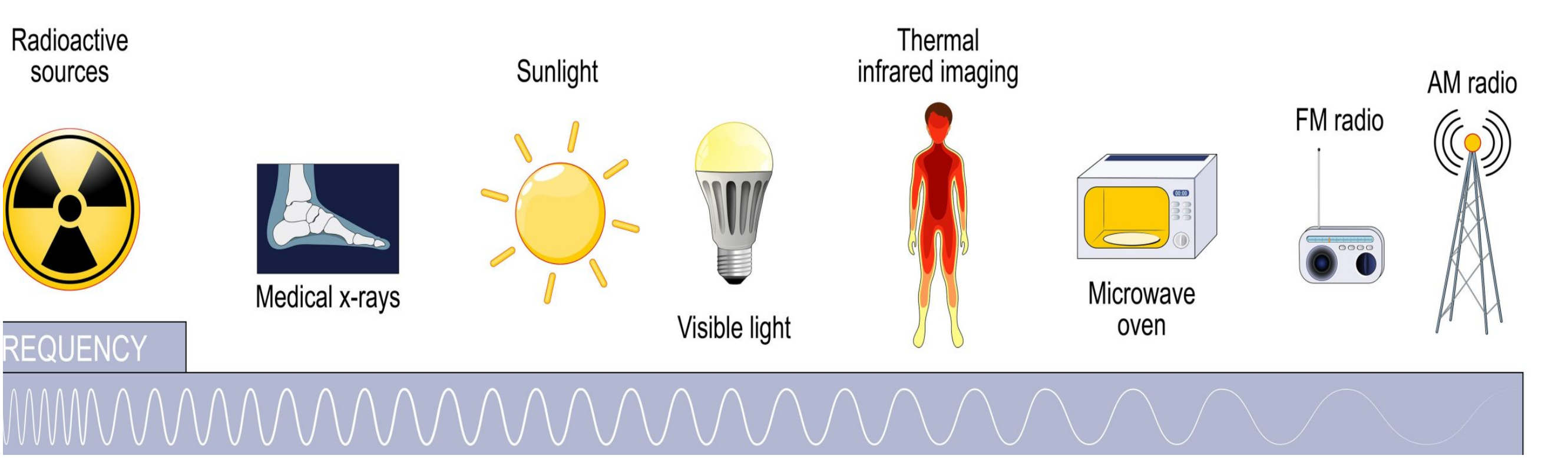
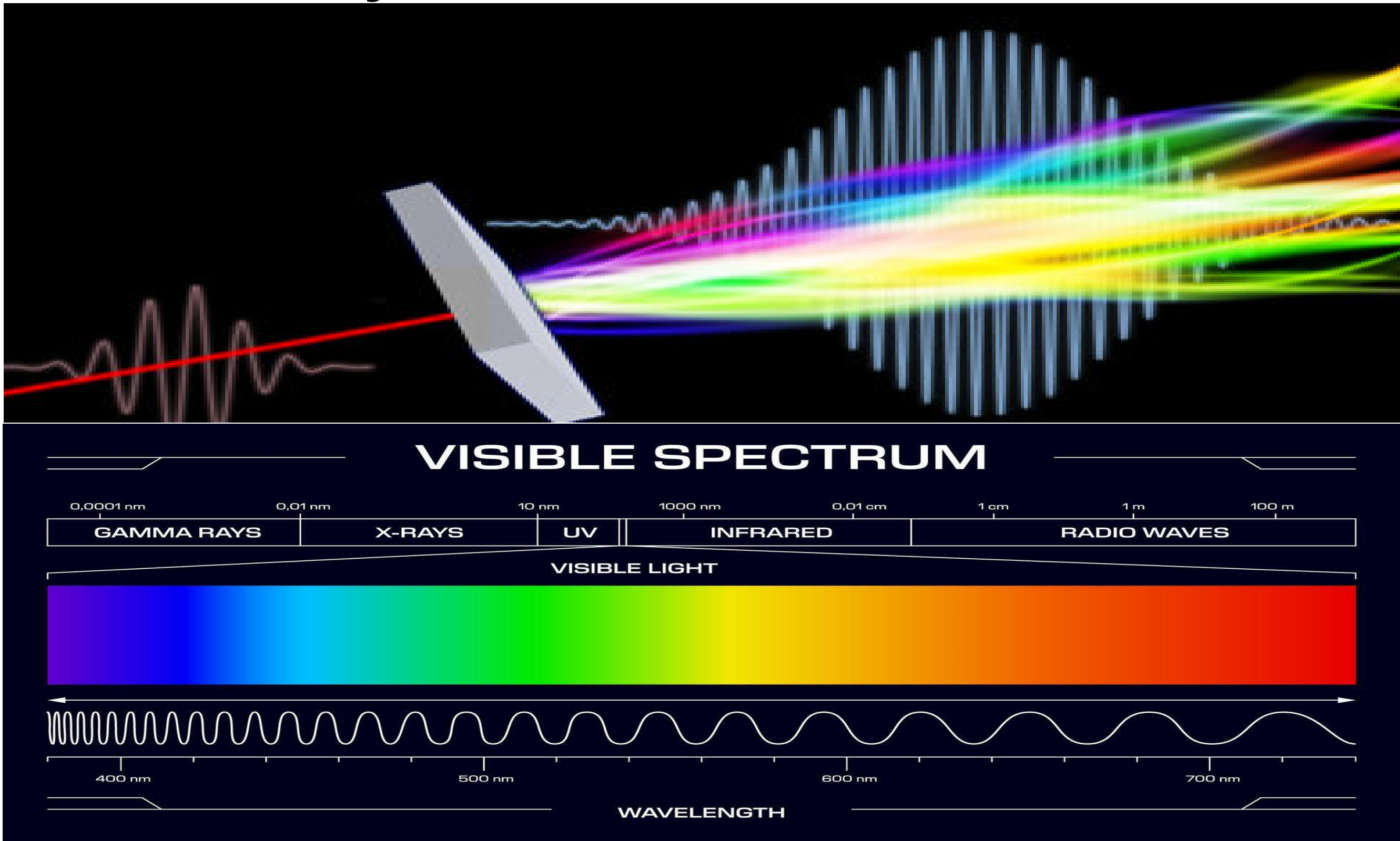
2. Multiphoton ionization



3. High Harmonic Generation



Have you ever seen a blue laser?



4. Interest

- From different ionizations can be obtained
 - electrons with even more energy (MPI)
 - a laser more energetic than the one we used (HHG)
 - electrons ionized even with insufficient energy (TI)
- Multiphoton Ionization (MPI)
 - electron is ionized by many photons instead of one
 - allows to ionize even with insufficient photon energy
- High Harmonic Generation (HHG)
 - allows to multiply laser energy manifold

5. More applications

- Quantum imagery technics (MRI, LIED) allowing to scan the body or the molecules
- High resolution camera allowing to track the movement of atoms at the femtosecond scale
- Attosecond physics in general...

